

# ADARSH ROUNIYAR

Kaohsiung, Taiwan ◇ +886 937 772 769 ◇ [adarsh@csie.io](mailto:adarsh@csie.io)  
[linkedin.com/in/adarsh-rouniyar](https://linkedin.com/in/adarsh-rouniyar) | [AI & GitHub Portfolio](#)

## SUMMARY

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AI & Web3 Systems Engineer building production-grade intelligent systems across **LLM platforms, RAG applications, multi-chain DeFi infrastructure, and security-focused developer tooling**. Proven record of shipping and operating end-to-end systems independently, including a live **multi-chain staking platform securing \$1M+ TVL**, self-hosted LLM infrastructure, AI-assisted engineering workflows, and real-time blockchain monitoring. Strong in **system ownership, production reliability, security hardening, and rapid execution from architecture to deployment**.

## SELECTED HIGHLIGHTS

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- Shipped and operated a **production multi-chain DeFi platform** securing **\$1M+ USDT/USDC TVL** across **BNB Chain, X Layer, and Base**.
- Architected a **secure referral and reward platform** with **EIP-191** signature verification, immutable wallet-code binding, admin-controlled claims, and anti-duplicate payout controls.
- Built **snapshot-based monthly reward accounting** with claimable-vs-paid reconciliation and auditable payout history for production settlement.
- Implemented **transaction reliability and event parsing logic** with RPC/indexer fallback to reduce false claim, unstake, and renewal failures in production UX.
- Built and deployed a **self-hosted multi-user LLM platform** with **custom routing, VS Code integration**, secure external access, and OpenAI-compatible APIs for internal engineering use.
- Built a **production LLM-powered crypto assistant** using **RAG + live market data** for contextual Q&A, analysis, and decision-support workflows.
- Building **Almond teAI**, an AI-powered code security scanner and trust platform with static analysis, secret detection, dependency checks, PDF reporting, and planned **smart contract audit workflows**.
- Delivered critical systems with **full ownership**, including platform security hardening, production monitoring, incident recovery, and live infrastructure operations.

## TECHNICAL SKILLS

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- **AI / LLM Systems:** LLM integration, RAG, prompt engineering, OpenAI-compatible APIs, llama.cpp, Ollama, VS Code Continue, Codex, Claude, multi-model orchestration, tool-augmented LLM systems
- **Programming:** Python, TypeScript, JavaScript, Rust, TensorFlow, PyTorch, Keras
- **Blockchain & Web3:** EVM smart contracts, multi-chain dApps (BSC / X Layer / Base), Wagmi, Ethers.js, wallet integration, EIP-191 signature verification, on-chain event handling
- **Backend & Security:** Node.js, serverless APIs, REST APIs, Firestore Admin SDK, authentication & session security, static code analysis, secret detection, dependency analysis, cron-based automation
- **Infrastructure:** Linux, Docker, Kubernetes (K8S), GCP, Cloudflare Tunnel, Git, Prisma, SQLite, Next.js, JupyterHub, Google Colab
- **ML / Research:** NLP pipelines, computer vision, federated learning, model aggregation, CNN-based systems

## PROFESSIONAL EXPERIENCE

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### Lead Backend & AI Engineer

Whale Strategy Co., Ltd., Kaohsiung, Taiwan

Apr 2024 – Present

- **WStaking Multi-Chain DeFi Platform (Live):** Led end-to-end delivery of a production staking platform and dashboard using **React + TypeScript**, integrating **Wagmi/Ethers** to route transactions across **BNB Chain, X Layer, and Base** with live staking, claim, renew, and unstake flows.
- **Smart Contract Integration & Production Operations:** Integrated and operated verified EVM contracts for staking, reward claims, auto-renew, and unstake flows across **X Layer, Base, and BSC**, securing **\$1M+ USDT/USDC TVL** on [wstaking.net](https://wstaking.net).
- **Referral Platform Architecture & Security:** Designed and implemented referral APIs for code generation/customization, wallet-code binding, nonce-based claim requests, and admin claim processing; enforced ownership verification with **EIP-191 signatures**, expiring challenge tokens, and server-side validation to prevent spoofing and replay abuse.
- **Referral Reward Accounting & Settlement:** Built monthly snapshot-based reward computation (**0.5%**) with claimable-vs-paid reconciliation, duplicate-claim blocking, scheduled cron snapshot jobs, and auditable payout history to support controlled referral settlement.
- **On-Chain Transaction Reliability:** Implemented robust receipt and event parsing for **Staked, RewardsClaimed, StakeRenewed, Unstaked, and PartialUnstaked**, including receipt normalization and fallback handling for RPC/indexer inconsistencies, improving production transaction reliability and reducing false error states.
- **Security Hardening & Monitoring:** Designed and deployed Telegram-based monitoring services on Linux servers to track real-time on-chain activity, staking events, wallet balance/allowance anomalies, unauthorized admin access attempts, and user connection analytics across three chains.
- **Platform Security & Audit Coordination:** Coordinated with smart contract audit firms to review and harden contracts, supported remediation and validation, and improved release readiness before production deployments.
- **Production Incident Response & Infrastructure Recovery:** Recovered critical company systems after a primary NVMe failure by performing sector-level data recovery, backup preservation, OS rebuild, environment restoration, and staged service recovery, protecting production data and restoring core bots/services.
- **Whale Strategy Internal AI Platform (Live):** Designed and deployed a self-hosted multi-user AI platform for internal engineering workflows, integrating **llama.cpp** and **Ollama** with a custom routing layer for coding and chat assistance. Enabled team-wide usage through **VS Code Continue** via **OpenAI-compatible APIs**. Built with **Docker**, exposed securely via **Cloudflare Tunnel**, and white-labeled for production use.
- **LLM Intelligence Systems:** Developed an LLM-powered crypto assistant leveraging **RAG** and live market data APIs for contextual Q&A, analysis, sentiment workflows, and decision-support use cases.
- **Quantitative Trading Infrastructure:** Built and containerized algorithmic trading bots using **Python, OKX APIs, and TradingView (Pine Script)** for automated live execution.
- **Web Operations:** Independently developed, secured, and hosted the company website: [whalestrategy.net](https://whalestrategy.net).
- **AI-Augmented Engineering Workflows:** Designed and adopted AI-assisted development pipelines using **Codex, Claude, and VS Code Continue**, integrating internal LLM systems via **OpenAI-compatible APIs** to support code generation, debugging, refactoring, and system design while maintaining production validation and safety practices.

**AI Research Assistant & Project Assistant**  
National Chung Cheng University (CCU), Taiwan

Nov 2021 – May 2024

- Conducted applied research in **federated learning** and **trustworthy AI** for sustainable smart-city systems, coordinating with international academic and government partners.
- Implemented **CNN-based prediction models** and distributed ML pipelines using TensorFlow and PyTorch; translated research outcomes into practical system designs.
- Authored and published research in **IEEE ISC2, ACM GoodIT, Future Internet, and Artificial Intelligence Review**.

- Contributed to the [NSTC Trustworthy AI Smart City Project](#).

## KEY PROJECTS

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- **Almond teAI — AI Code Security Scanner & Trust Platform (Live):** Built a full-stack security scanning platform using **Next.js, TypeScript, Prisma, and SQLite** to analyze GitHub repositories, uploaded codebases, and live websites. Implemented static code analysis, secret detection, dependency checks, website misconfiguration scanning, security scoring, regression tracking, PDF reporting, and public trust verification pages/badges. Extended the platform to support **smart contract audit workflows across EVM chains**, including multi-chain security analysis and trust reporting for Web3 projects.
- **WStaking Multi-Chain DeFi & Referral Platform (Live):** Built chain-aware staking flows and secure referral infrastructure including first-visit referral capture, signed bind/claim flow, snapshot-based reward accounting, and admin payout processing across BNB Chain, X Layer, and Base.
- **WS-AI Crypto Assistant (Production AI System) (Live):** Built LLM + RAG pipelines with real-time crypto market data for automated analysis and decision support, including structured market insights, sentiment workflows, and trading-oriented outputs.
- **Automated Trading Infrastructure:** Built containerized Python-based trading systems utilizing ML-assisted signal filtering and OKX execution. [GitHub Repo](#)
- **Eff-AQI Federated Learning System:** Built a CNN-based air pollution estimation system published at ACM GoodIT. Dataset: [GitHub](#) | [Kaggle DOI](#)

## EDUCATION

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**M.Sc. Computer Science and Information Engineering** 2021 – 2023  
National Chung Cheng University, Taiwan GPA: 4.11 / 4.3

**B.Tech. Computer Science and Engineering** 2017 – 2021  
Vel Tech University, India CGPA: 8.3 / 10

## SELECTED PUBLICATIONS

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- **Federated Learning Design and Functional Models: Survey.** *Artificial Intelligence Review*, 2024. [DOI](#)
- **Federated Adversarial Training Strategies for Sustainable Smart Cities.** *Future Internet*, 2023. [DOI](#)
- **Eff-AQI: Efficient CNN-Based Air Pollution Estimation.** *ACM GoodIT*, 2023. [DOI](#)
- **Federated Trustworthy AI Architecture for Smart Cities.** *IEEE ISC2*, 2022. [DOI](#)

## ADDITIONAL

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- **Languages:** English, Hindi, Nepali; Mandarin (basic)
- **Live Systems:** [WStaking Platform](#) | [Internal AI Platform](#) | [WS-AI Crypto Assistant](#) | [Almond teAI](#)
- **Code & Portfolio:** [Trading Systems Repo](#) | [Personal Portfolio](#)